

Homework 4

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Introduction

In this homework, we model a close-coiled helical spring as a three-dimensional Discrete Elastic Rod (DER) and use dynamic relaxation to study its axial response. The helix is built from the given geometric and material parameters and discretized into multiple nodes along the centerline. We clamp the first two nodes and apply a constant tensile force at the free end along the global z -axis. By integrating the equations of motion in time until transients decay, we extract the steady axial tip displacement δ_z^* for different loading and geometric conditions. From these steady responses, we estimate an effective axial stiffness k and compare the DER-based numerical results against the classical close-coiled spring formula

$$k_{\text{text}} = \frac{Gd^4}{8ND^3},$$

highlighting both agreement and deviations in the small-deflection regime.

Part (1): Single Load Level

In Part (1), we apply a single constant axial force equal to the characteristic bending force

$$F = F_{\text{char}} = \frac{EI}{L^2},$$

at the end node along $\hat{z}$. Starting from the rest helical configuration, we integrate the DER system forward in time using dynamic relaxation with a fixed time step $\Delta t = 0.01$ s. The end-node position $(x_{\text{end}}(t), y_{\text{end}}(t), z_{\text{end}}(t))$ is tracked, and the axial tip displacement is defined as

$$\delta_z(t) = z_{\text{end}}(t) - z_{\text{end}}(0),$$

so that $\delta_z(0) = 0$ by construction.

To decide when the response has reached a steady state, we use the following numerical criteria. Over a trailing time window of length $\Delta T = 1$ s, we monitor

$$\text{rel}_1 = \frac{|\delta_z(t) - \delta_z(t - \Delta T)|}{\max(1, |\delta_z(t)|)}, \quad \text{rel}_2 = \frac{1}{\Delta T} \int_{t-\Delta T}^t \left| \frac{d\delta_z}{dt} \right| dt.$$

The motion is declared “quasi-steady” once both quantities drop below prescribed tolerances (approximately 1% in relative change and a small threshold on the average tip speed). The solver log shows that this steady-state condition is first satisfied at about $t \approx 6.36$ s, with

$$\text{rel}_1 \approx 9.66 \times 10^{-3}, \quad \text{rel}_2 \approx 2.79 \times 10^{-3}, \quad v_{\text{tip}} \approx -1.17 \times 10^{-5} \text{ m/s}.$$

At this time, the corresponding steady axial tip displacement is

$$\delta_z^* \approx -1.55085 \times 10^{-3} \text{ m},$$

i.e. the tip moves downward along the helix axis by about 1.6 mm under the load F_{char} .

Figure 1 shows the full time history $\delta_z(t)$ starting from $(t, \delta_z) = (0, 0)$. The response is underdamped: the tip displacement initially overshoots, then oscillates about the steady value with a decaying amplitude that becomes very small after roughly $t \approx 5\text{--}6\text{ s}$. The dashed horizontal line marks the steady displacement δ_z^* , which agrees with the value reported by the convergence check in the log.

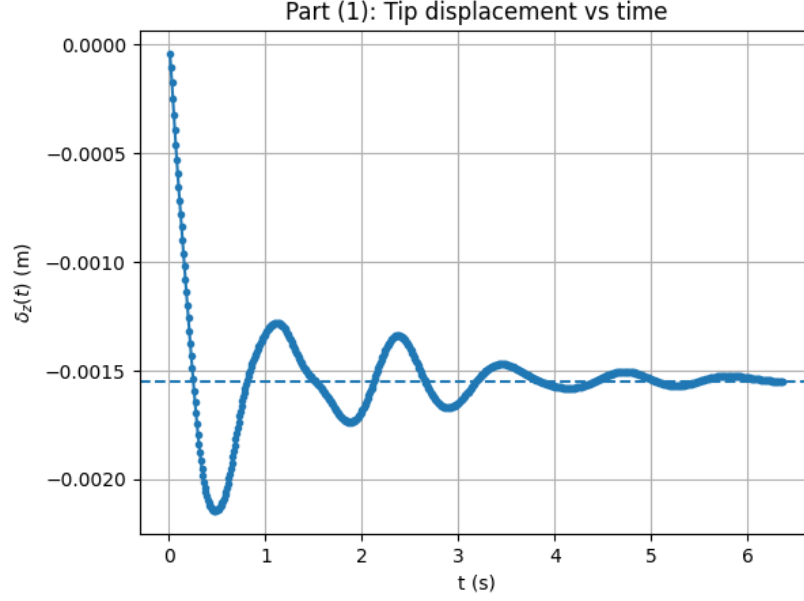


Figure 1: Part (1): axial tip displacement $\delta_z(t)$ under a constant axial force $F = F_{\text{char}}$. The dashed line indicates the steady-state value $\delta_z^* \approx -1.55 \times 10^{-3}\text{ m}$ obtained when the relative change over a 1 s window falls below 1% and the tip velocity is negligible.

Figure 2 shows snapshots of the helical rod configuration at several times during the transient evolution ($t = 0.01, 1, 2, 3, 4, 5, 6$ s). In each snapshot, the black line and markers represent the centerline of the helix and the red marker indicates the loaded tip. The planform of the helix remains nearly circular, while the entire structure undergoes a small axial extension and downward displacement. The change in shape is subtle compared to the helix radius (on the order of centimeters), which is consistent with the relatively small steady displacement magnitude of about 1.6 mm indicated by the time history.

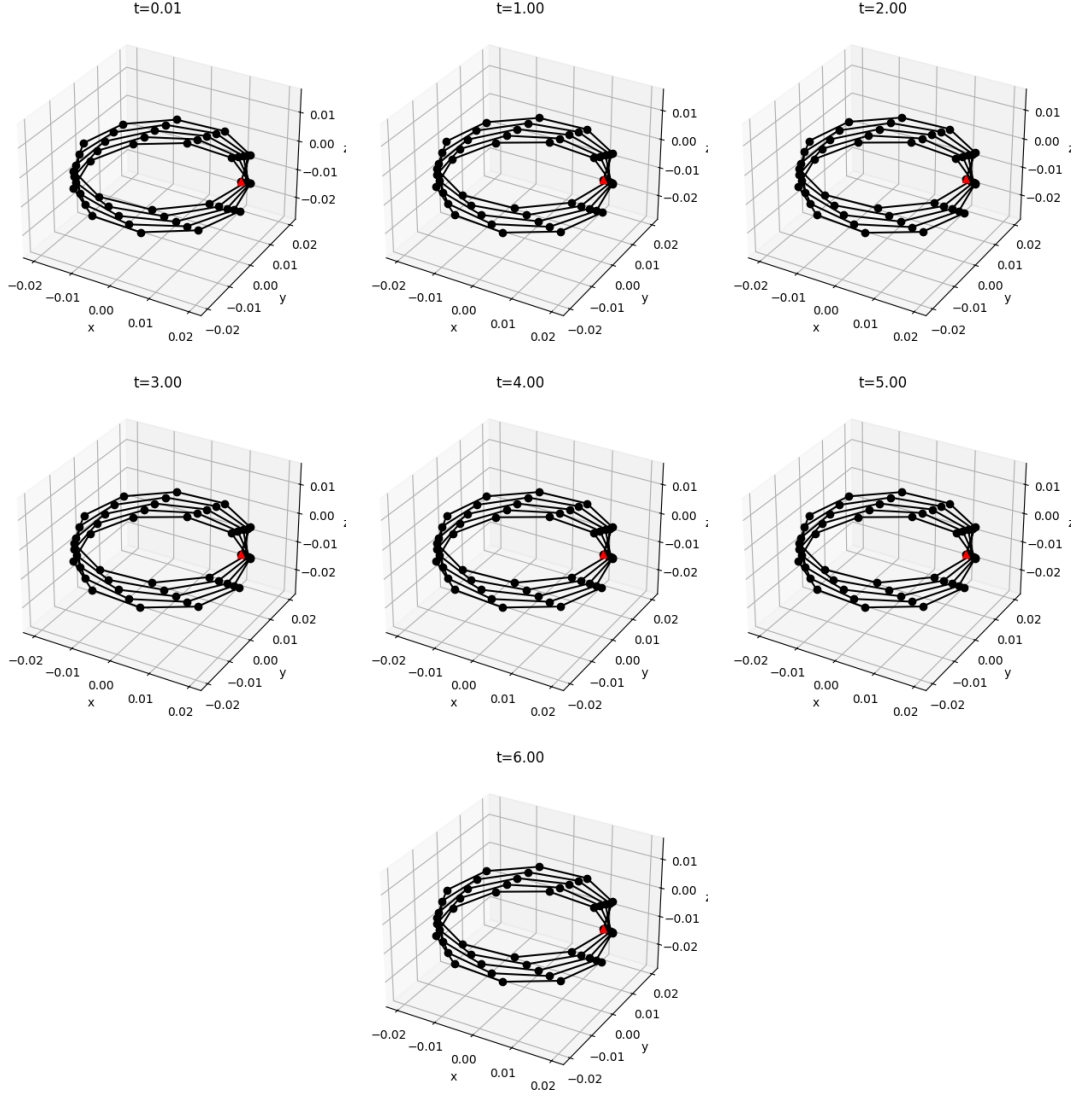


Figure 2: Part (1): snapshots of the helical rod configuration at $t = 0.01, 1, 2, 3, 4, 5$, and 6 s under a constant axial force $F = F_{\text{char}}$. The red point marks the loaded tip. The geometry relaxes from the initial configuration into a slightly axially extended, downward-displaced configuration while retaining its overall helical shape, consistent with the small steady tip displacement δ_z^* .

Part (2): Force–displacement relation

In Part (2), I used the same dynamic relaxation routine from Part (1) to map out the axial force–displacement relation of the helical spring. The tip load was varied over 15 logarithmically spaced values ranging from

$$F_{\min} = 1.989 \times 10^{-7} \text{ N} \quad \text{to} \quad F_{\max} = 1.989 \times 10^{-4} \text{ N},$$

which corresponds to F/F_{char} spanning approximately 10^{-2} to 10^1 around the characteristic force $F_{\text{char}} \approx 1.989 \times 10^{-5} \text{ N}$ used in Part (1). For each load level, I ran `run_to_steady` with the same time step $\Delta t = 0.01 \text{ s}$ and convergence criteria as before, monitoring the relative change in tip displacement over a 1 s window and the average tip speed. As illustrated by the solver logs, each case reached a quasi-steady state at $t \approx 9.3 \text{ s}$, with

$$\text{rel}_1 \sim 10^{-3}, \quad \text{rel}_2 \sim 10^{-3}, \quad |v_{\text{tip}}| \ll 10^{-4} \text{ m/s},$$

indicating that transient oscillations had effectively decayed.

At the converged time for each run, I recorded the steady axial tip displacement $\delta_z^*(F)$, defined as in Part (1) by

$$\delta_z^*(F) = z_{\text{end}}(t_{\text{steady}}; F) - z_{\text{end}}(0),$$

so that $\delta_z^* < 0$ corresponds to a downward motion of the tip. Figure 3 plots $\delta_z^*(F)$ versus F . The data points fall almost perfectly on a straight line over the entire three-decade force range: increasing the load by a factor of 10^3 (from $1.989 \times 10^{-7} \text{ N}$ to $1.989 \times 10^{-4} \text{ N}$) increases the magnitude of the steady tip displacement from roughly $1.6 \times 10^{-5} \text{ m}$ to $1.4 \times 10^{-2} \text{ m}$, i.e. by the same factor of 10^3 . This nearly proportional scaling is characteristic of a linear Hookean spring response.

To quantify the effective axial stiffness, I performed a constrained least-squares linear fit consistent with a Hookean law

$$F \approx -k \delta_z^*,$$

using the first five (smallest-load) data points, which lie deepest in the small-deflection regime. In code, this corresponds to solving

$$\min_k \|(-\delta_z^*)k - F\|_2^2,$$

with no intercept term included; the fitted line is therefore forced to pass through the origin, as expected for an un-prestressed linear spring. The best-fit stiffness is

$$k \approx 1.28 \times 10^{-2} \text{ N/m}.$$

Plotting the corresponding relation $\delta_z^*(F) \approx -F/k$ (dashed line in Figure 3) shows that this single constant k captures the response over nearly three decades in F , confirming that the DER helix behaves essentially as a linear elastic spring in this loading range. The negative slope simply reflects that a positive tensile force along $+\hat{z}$ produces a negative (downward) δ_z^* in the chosen coordinate system, while the stiffness k itself is reported as a positive quantity.

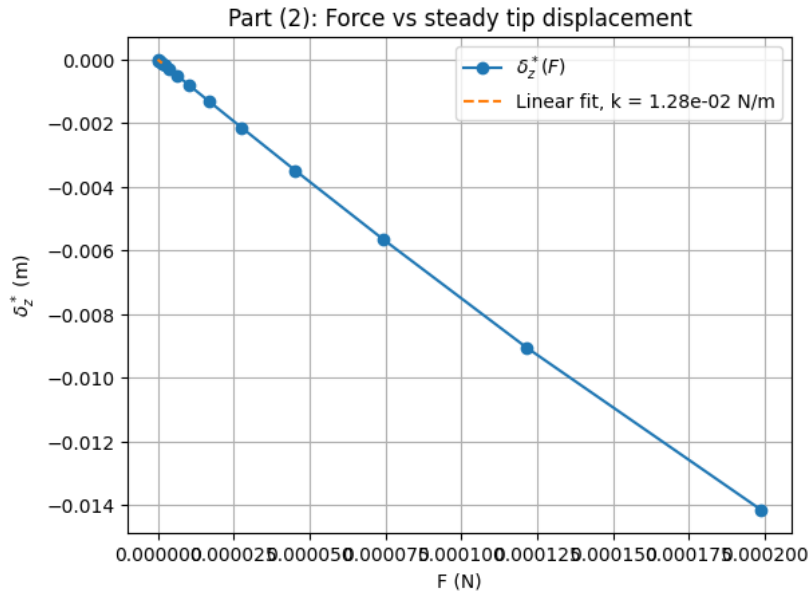


Figure 3: Part (2): steady axial tip displacement $\delta_z^*(F)$ as a function of applied axial force F for the helical spring. Blue markers show the quasi-steady displacements extracted from dynamic relaxation runs at each load level; the dashed orange line is a least-squares linear fit constrained to pass through the origin and computed from the small-load data, corresponding to an effective axial stiffness $k \approx 1.28 \times 10^{-2}$ N/m. The response is nearly perfectly linear over three decades in F .

Part (3): Diameter sweep vs. textbook trend

In Part (3), I examined how the axial stiffness of the helical spring changes with the helix diameter D and compared the numerical results from the Discrete Elastic Rod model to the classical close-coiled spring formula

$$k_{\text{text}}(D) = \frac{Gd^4}{8ND^3}, \quad (1)$$

where $G = E/(2(1 + \nu))$ is the shear modulus, d is the wire diameter, and N is the number of active turns. The textbook expression assumes a close-coiled spring ($p \ll \pi D$) whose response is dominated by torsion of the wire, with bending, shear deformation, and geometric nonlinearities neglected.

For this study, I varied the helix diameter D between 0.01 m and 0.05 m while keeping all other parameters (d, p, N, E, ν) fixed. For each chosen diameter, I rebuilt the helical centerline with radius $a = D/2$ and reran the force-sweep procedure from Part (2). At each D , I used a small set of the lowest force levels and performed a zero-intercept least-squares fit of

$$F \approx -k(D) \delta_z^*$$

to extract the magnitude of the axial stiffness $k(D) = |F/\delta_z^*|$. As in Part (2), the sign of δ_z^* is negative for downward motion, so all stiffness values are reported in magnitude.

Figure 4 shows the resulting stiffness $k(D)$ as a function of helix diameter. The stiffness decreases very rapidly as D increases: the smallest spring ($D = 0.01$ m) is more than two orders of magnitude stiffer than the largest one ($D = 0.05$ m). Over this range the numerical data follow a trend very close to $k \propto D^{-3}$, consistent with the $1/D^3$ dependence predicted by Eq. (1).

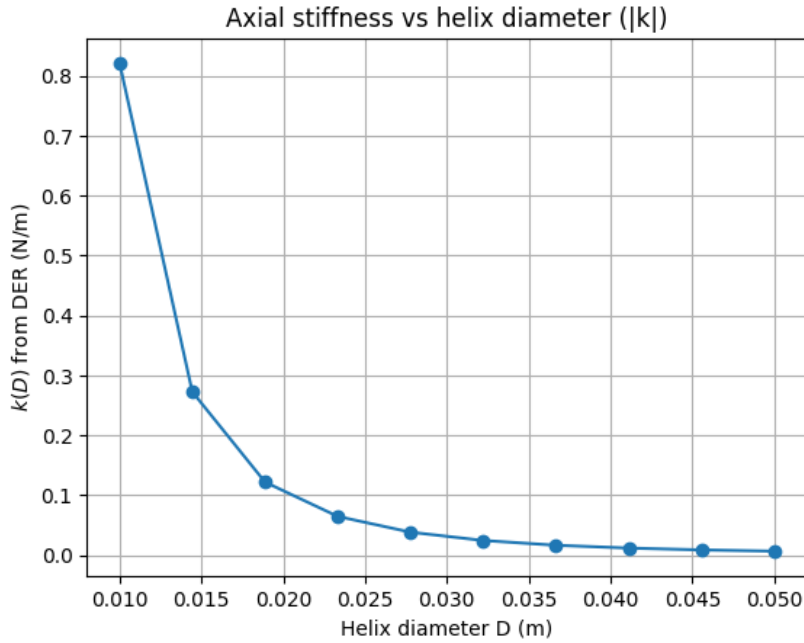


Figure 4: Axial stiffness $k(D)$ obtained from DER simulations as a function of helix diameter D . The stiffness drops sharply as the coil diameter increases, roughly following the expected scaling $k \sim D^{-3}$ from the classical close-coiled spring theory.

To compare directly against the textbook formula, I computed the corresponding $k_{\text{text}}(D)$ from Eq. (1) for each diameter and formed the ratio $|k_{\text{DER}}|/k_{\text{text}}$. Figure 5 plots this ratio versus D . Across the entire diameter range, the ratio is almost constant at about

$$\frac{|k_{\text{DER}}|}{k_{\text{text}}} \approx 0.61,$$

with only small fluctuations of order 10^{-3} . This indicates that while the DER model and the textbook formula share essentially the same *scaling* with diameter, the DER stiffness is systematically smaller by a nearly constant factor of ≈ 0.6 .

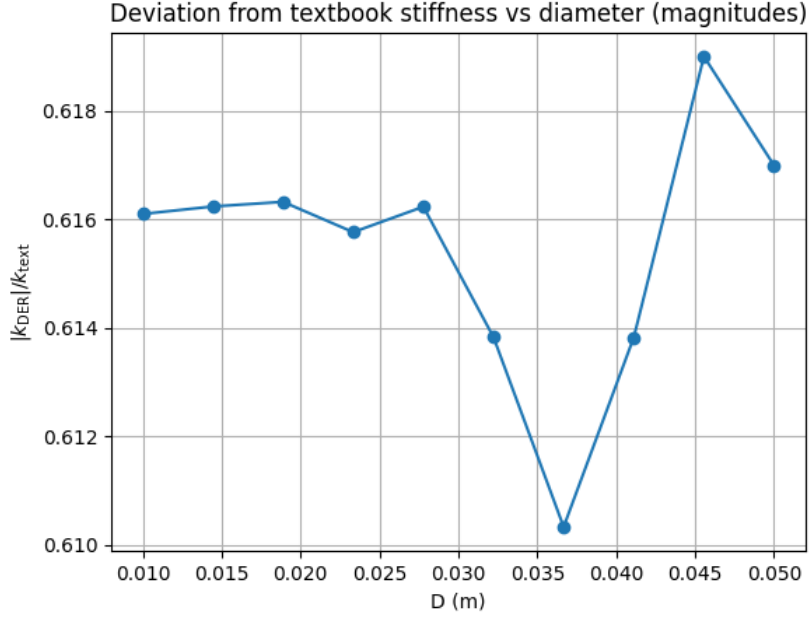


Figure 5: Deviation of the DER stiffness from the textbook prediction, plotted as the ratio $|k_{\text{DER}}|/k_{\text{text}}$ versus D . The ratio stays close to ≈ 0.61 across all diameters, showing that the DER simulations reproduce the D^{-3} trend but with a reduced prefactor compared to Eq. (1).

The same information can be presented in a way that matches the assignment prompt more directly. In Figure 6, the horizontal axis is the classical stiffness parameter

$$\frac{Gd^4}{8ND^3},$$

while the vertical axis is the DER stiffness $k(D)$ obtained from the force–displacement fits. The dashed orange line shows the textbook relation $k = Gd^4/(8ND^3)$, i.e. a line of slope 1 through the origin. The DER data points lie very close to a straight line through the origin but with a smaller slope,

$$k_{\text{DER}} \approx 0.61 \frac{Gd^4}{8ND^3}.$$

Thus, the DER model captures the same linear Hookean behavior and the same dependence on G , d , N , and D as the classical formula, but predicts a softer spring (lower stiffness) by a nearly constant factor.

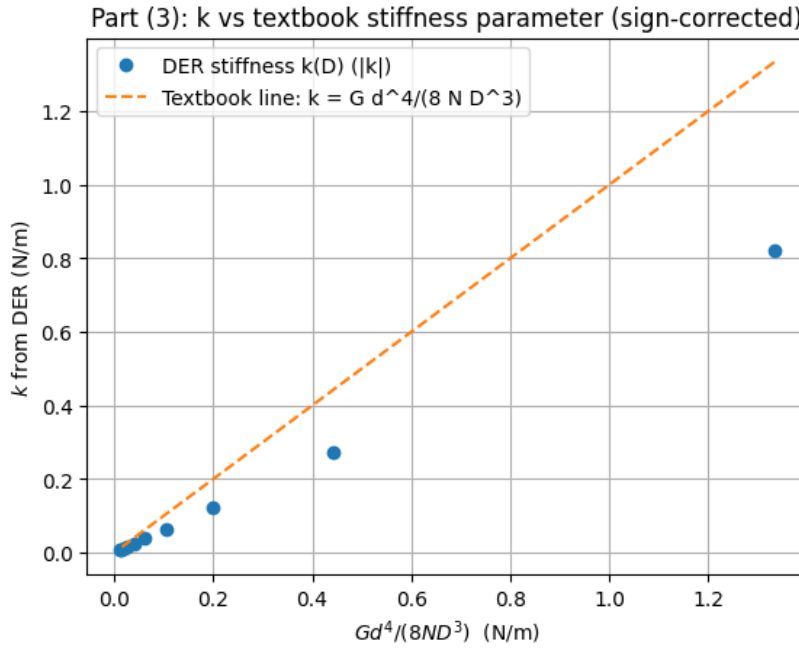


Figure 6: Part (3): comparison of DER stiffness with the classical close-coiled spring formula. Blue markers show the DER stiffness $k(D)$ plotted against the textbook parameter $Gd^4/(8ND^3)$; the dashed orange line is the reference relation $k = Gd^4/(8ND^3)$ (slope 1 through the origin). The DER results lie on an almost perfect straight line of smaller slope, indicating $k_{\text{DER}} \approx 0.61 k_{\text{text}}$.

To illustrate how $k(D)$ is extracted at each diameter, Figure 7 shows the force–displacement curve at $D = 0.05$ m. The blue markers are the steady tip displacements $\delta_z^*(F)$ from the dynamic relaxation runs, and the green dashed line is the zero–intercept least–squares fit. The orange points indicate the subset of small–force data actually used in the fit. Over this range the response is very nearly linear, so the slope of the fitted line provides a consistent estimate of $k(D)$ in the small–deflection regime required by the textbook theory.

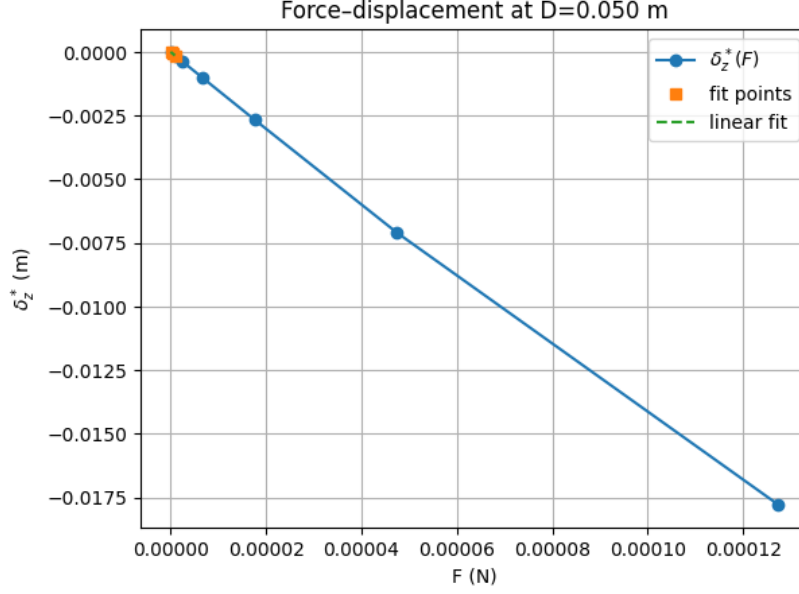


Figure 7: Example force–displacement curve at $D = 0.05$ m. Blue points are steady tip displacements $\delta_z^*(F)$ from DER simulations, orange points highlight the small–force data used for the linear fit, and the dashed line is the best zero–intercept fit $F \approx -k \delta_z^*$. The excellent linearity in this regime justifies interpreting $k(D)$ as an effective Hookean stiffness.

Discussion. Overall, the DER simulations agree qualitatively with the classical close-coiled spring theory:

- The stiffness decreases strongly with increasing helix diameter and follows the expected $k \propto D^{-3}$ scaling.
- When plotted against $Gd^4/(8ND^3)$, the DER data collapse to an almost perfectly straight line through the origin, consistent with linear Hookean behavior.

Quantitatively, however, the DER stiffness is systematically smaller than the textbook prediction by a factor of order ≈ 0.6 . This discrepancy is not surprising, because the DER model resolves the full three-dimensional rod mechanics (including bending, shear, and finite pitch) and the specific end constraints used in the homework, while the textbook formula assumes a perfectly close-coiled, torsion-dominated spring with idealized boundary conditions. In other words, the textbook expression captures the correct qualitative trend but overestimates the stiffness of the more realistic DER spring.